Scientific Validation Plan for the Bright–Dark Model Used to Probe the Spinorial Sector of a Spin-Peierls Chain

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1 Purpose of the plan

The goal is not only to produce spectra. The goal is to build a sequence of tests that can answer, in a falsifiable way, the central question:

> Can multidimensional optical spectroscopy, through an effective spin-orbit coupling, probe the dynamics of the spin sector in a Spin-Peierls chain?

The current model is an effective model. It does not yet contain a complete microscopic Hamiltonian with d orbitals, crystal field, explicit spin-orbit coupling, phonons, and a dynamic spin chain. The strategy must therefore separate three levels:

1. **What the solver actually computes**: a multidimensional optical response of a bright–dark–mode Hamiltonian.
2. **What the parameters encode**: δ, C_1, V_{BD}, $\omega_Q(k)$, $g_Q(k)$.
3. **What we want to infer physically**: spin correlations, Spin-Peierls dynamics, magnetic-field effects, bright–dark coherence, and the role of SOC.

The strategy is to add one physical layer at a time. Each layer must have a control in which the expected signature disappears. If a signature is already present in a simpler model, it cannot be assigned unambiguously to the more complex mechanism.

2 Reference model to validate

The target effective Hamiltonian is

$$H(k) = E_D \left|D\right\rangle \left\langle D\right| + E_B \left|B\right\rangle \left\langle B\right| + \omega_Q(k)a^\dagger a + V_{\mathrm{BD}}^0 L_{\mathrm{BD}} + g_Q(k)L_{\mathrm{BD}}(a + a^\dagger), \tag{1}$$

with

$$L_{\mathrm{BD}} = \left|D\right\rangle \left\langle B\right| + \left|B\right\rangle \left\langle D\right|, \tag{2}$$

and

$$V_{\mathrm{BD}}^0 = V_0 + \lambda_\delta \delta + \lambda_C C_1. \tag{3}$$

The dispersive part is

$$\omega_Q(k) = \omega_Q + s\left[\omega_{\mathrm{SP}}(k) - \omega_{\mathrm{SP}}(0)\right], \tag{4}$$

and the dynamic coupling is

$$g_Q(k) = g_Q\left(1 + A_Q \cos k\right). \tag{5}$$

The model should be interpreted as an inference chain:

$$\text{spin chain} \to (\delta, C_1, \Delta_{\text{spin}}) \to (V_{\text{BD}}, \omega_Q(k), g_Q(k)) \to S(\omega_1, \omega_3). \tag{6}$$

3 Scientific questions and target observables

ID	Scientific question	Main observable	Essential control
Q1	Can multidimensional spectroscopy probe the spin sector through SOC?	Variation of $S(\omega_1, \omega_3)$, cross peaks, integrated intensities, phases.	Turn off the spin–optical link: $\lambda_\delta = \lambda_C = g_Q = 0$.
Q2	Is bright–dark coherence an indicator of quantum correlations?	Amplitude and phase of $B \leftrightarrow D$ cross peaks, oscillations in t_2.	Compare a model with fixed C_1 to a model where $C_1(T, B)$ varies.
Q3	Are cross peaks a signature of coherent eigenstate mixing?	Scaling of cross peaks with V_{BD}, eigenvector analysis.	$\mu_D = 0$, $V_{\text{BD}} = 0$: the cross peak must disappear.
Q4	Which microscopic quantity does the cross-peak amplitude measure?	Derivatives $\partial I/\partial \delta$, $\partial I/\partial C_1$, $\partial I/\partial g_Q$.	Independent scans of δ, C_1, g_Q.
Q5	Does the magnetic field tune coherent mixing or only shift energies?	Peak shifts versus intensity changes.	Compare field dependence in H_{spin}, in $T_{\text{SP}}(B)$, and in the optical energies separately.
Q6	Can nonlinear spectroscopy distinguish thermal and nonthermal states?	Spectra computed from thermal populations versus imposed populations.	Same Hamiltonian; only the initial distribution changes.
Q7	Do cross peaks encode short-range spin correlations?	Correlation between I_{cross} and C_1.	Keep δ fixed and vary only C_1, then do the inverse.
Q8	What is the role of detection in accessing the dark state?	Comparison of excitation/detection through μ_B, μ_D, and eigenstate mixing.	Compare $\mu_D = 0$ and $\mu_D \neq 0$.
Q9	Are the observed correlations robust against numerical artifacts?	Convergence in N_k, η, frequency grid, normalization, chain size.	Reproduce the trends at several resolutions.

4 General validation principles

4.1 Add ingredients one by one

The recommended validation order is:

$$\begin{gathered}\text{minimal bright/dark} \to \text{bosonic mode} \to \text{spin observables} \\ \to k\text{-dispersion} \to \text{field/temperature} \to \text{explicit SOC}.\end{gathered} \tag{7}$$

Each step must keep the previous-step results as controls. A new signature is interpretable only if it is absent in the corresponding control.

4.2 Always keep non-normalized spectra

Panel-normalized figures are useful for comparing shapes. They do not allow conclusions about absolute intensity. For each spectrum, one should save at least

$$S_{R1}^{\text{real}}, \quad S_{R1}^{\text{imag}}, \quad |S_{R1}|, \quad S_{R2}^{\text{real}}, \quad S_{R2}^{\text{imag}}, \quad |S_{R2}|, \tag{8}$$

with and without normalization.

4.3 Measure scalars, not only contour plots

The minimal observables are:

$$I_{\max} = \max_{\omega_1,\omega_3} |S(\omega_1,\omega_3)|, \tag{9}$$

$$I_{\text{int}} = \int d\omega_1\, d\omega_3\, |S(\omega_1,\omega_3)|, \tag{10}$$

$$I_{\text{cross}} = \int_{\Omega_{\text{cross}}} d\omega_1\, d\omega_3\, |S(\omega_1,\omega_3)|, \tag{11}$$

$$I_{\text{diag}} = \int_{\Omega_{\text{diag}}} d\omega_1\, d\omega_3\, |S(\omega_1,\omega_3)|, \tag{12}$$

$$R_{\text{cross}} = I_{\text{cross}}/I_{\text{diag}}. \tag{13}$$

The peak positions should also be saved:

$$(\omega_1^*, \omega_3^*) = \arg\max_{\omega_1,\omega_3} |S(\omega_1,\omega_3)|. \tag{14}$$

5 Step 0 – Numerical hygiene and conventions

5.1 Objective

Before interpreting the physics, one must verify that the numerical conventions do not create the signatures. This step does not yet answer the physical question; it prevents building an interpretation on an artifact.

5.2 Tests

Test point. T0.1 – Normalization control. For the same parameter set, produce spectra with global normalization, panel normalization, and no normalization. The shapes may look similar, but conclusions about intensity must be drawn only from non-normalized spectra.

Test point. T0.2 – Frequency-grid convergence. Recompute a representative spectrum with at least three grid resolutions. Peak positions and integrals over fixed windows must converge.

Test point. T0.3 – Broadening convergence in η. Test several values of η. A robust physical signature should deform continuously with η, not appear or disappear discontinuously.

Test point. T0.4 – $R1$ and $R2$ pathway control. Save $R1$, $R2$, and their sum separately. A variation assigned to physics must be localizable: either it affects both pathways coherently, or it reveals an interpretable asymmetry.

Decision criterion. Proceed to Step 1 only if the non-normalized spectra, peak positions, and integrated intensities are numerically stable for the minimal model.

6 Step 1 – Minimal optical system without mode and without spin

6.1 Minimal system

The first system should be as small as possible:

$$\mathcal{H}_0 = \mathrm{span}\{|g\rangle, |B\rangle, |D\rangle\}. \tag{15}$$

Impose:

$$N_k = 1, \qquad g_Q = 0, \qquad \omega_Q \text{ absent}, \qquad \mu_D = 0. \tag{16}$$

The optical Hamiltonian is

$$H_0 = E_B |B\rangle\langle B| + E_D |D\rangle\langle D| + V_{\mathrm{BD}} L_{\mathrm{BD}}. \tag{17}$$

6.2 Why start here

This system isolates the most fundamental mechanism: does a dark state become visible because it is mixed with a bright state? If this step is not understood, scans in δ, C_1, k, and $g_Q(k)$ will be ambiguous.

6.3 Tests

Test point. T1.1 – Strictly invisible dark state. Set $\mu_D = 0$ and $V_{\mathrm{BD}} = 0$. The spectrum should contain only signatures associated with $|B\rangle$. Any signature of $|D\rangle$ in this control indicates a problem with the dipole, the basis, the spectral window, or the optical pathway.

Test point. T1.2 – Controlled appearance of the dark peak. Scan V_{BD} on a logarithmic scale, for example from 10^{-5} to 10^{-1} eV if numerically reasonable. Measure I_D, I_{cross}, and R_{cross}.

Test point. T1.3 – Mixing scaling law. For $|V_{\mathrm{BD}}| \ll |E_D - E_B|$, the bright weight of the dark state should have the order of magnitude

$$\left|\frac{V_{\mathrm{BD}}}{E_D - E_B}\right|^2. \tag{18}$$

The log–log slope of the dark intensity or cross-peak intensity should therefore be measured empirically. Depending on the optical pathway and the chosen observable, the slope can be linear or quadratic in V_{BD}; it should be documented rather than assumed.

Test point. T1.4 – Detuning dependence. Repeat the scan for several $\Delta_{BD} = E_D - E_B$. If the dark signal truly comes from mixing, it should increase when $|\Delta_{BD}|$ decreases, all else being equal.

Test point. T1.5 – Sign of V_{BD}. Compare $+V_{\mathrm{BD}}$ and $-V_{\mathrm{BD}}$. Absolute intensities may be nearly identical, but real/imaginary parts and phases can change. This comparison helps distinguish an intensity signature from a coherence signature.

Decision criterion. This step partially answers Q3. If the cross peaks disappear when $V_{\mathrm{BD}} = 0$ and grow systematically with V_{BD}, then they are compatible with coherent bright–dark mixing. Otherwise, they cannot be interpreted as evidence of SOC or spin correlations.

7 Step 2 – Add a local mode without k dependence

7.1 System

Add a truncated mode:

$$\mathcal{H}_1 = \mathrm{span}\{|g\rangle, |B\rangle, |D\rangle\} \otimes \mathcal{H}_{\mathrm{ph}}, \tag{19}$$

but keep:

$$N_k = 1, \qquad \omega_Q(k) = \omega_Q, \qquad g_Q(k) = g_Q. \tag{20}$$

The Hamiltonian becomes:

$$H_1 = E_B |B\rangle\langle B| + E_D |D\rangle\langle D| + \omega_Q a^\dagger a + V_{\mathrm{BD}} L_{\mathrm{BD}} + g_Q L_{\mathrm{BD}}(a + a^\dagger). \tag{21}$$

7.2 Objective

This step separates static mixing V_{BD} from mode-assisted dynamic mixing g_Q. It indicates whether spectral structures come from simple bright–dark hybridization or from coupling to a collective coordinate.

7.3 Tests

Test point. T2.1 – Control without dynamic mode. Set $g_Q = 0$ and compare to Step 1. Differences should come only from the larger Hilbert space, not from a new physical coupling.

Test point. T2.2 – Activation of g_Q. Scan g_Q at fixed V_{BD}. Measure the appearance of satellites, peak deformation, and phase changes in $R1$ and $R2$.

Test point. T2.3 – Control with $V_{\mathrm{BD}} = 0$, $g_Q \neq 0$. This test is important: if the dynamic mode alone can produce a dark signal, then dark-state access does not depend only on static mixing. If the signal disappears, the mode mainly acts as a modulation of already-present mixing.

Test point. T2.4 – Bosonic truncation convergence. Repeat the test for $n_{\mathrm{bosons}} = 2, 3, 4, \ldots$. Intensities and peak positions should converge in the energy window used.

Decision criterion. This step answers the difference between static mixing and dynamic coupling. One should not introduce k before identifying what g_Q already does in a single-mode model.

8 Step 3 – Connect δ and C_1 without dispersion

8.1 System

Keep $N_k = 1$ and replace the free parameter V_{BD} by

$$V_{\mathrm{BD}} = V_0 + \lambda_\delta \delta + \lambda_C C_1. \tag{22}$$

At first, δ and C_1 should be controlled manually, without yet using the full spin chain.

8.2 Objective

This step directly addresses Q4 and Q7: does the spectrum measure δ, C_1, or only the effective combination V_{BD}?

8.3 Tests

Test point. T3.1 – Scan δ only. Set $C_1 = C_1^{\mathrm{ref}}$, $\lambda_C = 0$, and scan δ. Measure $I_{\mathrm{cross}}(\delta)$, $I_D(\delta)$, and peak positions.

Test point. T3.2 – Scan C_1 only. Set $\delta = \delta^{\mathrm{ref}}$, $\lambda_\delta = 0$, and scan C_1. Measure the same observables.

Test point. T3.3 – Parameter degeneracy. Compare two pairs (δ, C_1) that give the same V_{BD}. If the spectra are identical, the optical model does not distinguish δ and C_1; it measures only V_{BD}.

Test point. T3.4 – Local sensitivities. Compute numerically:

$$\frac{\partial I_{\mathrm{cross}}}{\partial \delta}, \qquad \frac{\partial I_{\mathrm{cross}}}{\partial C_1}, \qquad \frac{\partial \phi_{\mathrm{cross}}}{\partial \delta}, \qquad \frac{\partial \phi_{\mathrm{cross}}}{\partial C_1}. \tag{23}$$

Phases may contain information different from intensity.

Decision criterion. If two pairs (δ, C_1) with the same V_{BD} produce the same spectrum, then spectroscopy does not separately measure δ and C_1 in this model. To separate them, one will need to add a channel where δ and C_1 affect different terms, for example $\omega_Q(k)$, $g_Q(k)$, or dephasing rates.

9 Step 4 – Introduce the finite spin chain

9.1 System

Now compute $\delta(T, B)$, $C_1(T, B)$, and Δ_{spin} from the spin sector:

$$H_{\text{spin}} = \sum_i J_i \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} + J_2 \sum_i \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+2} + g\mu_B B \sum_i S_i^z. \tag{24}$$

9.2 Objective

The objective is to verify that the temperature/field dependence of the spectrum truly comes from spin observables, not from an arbitrary variation of optical parameters.

9.3 Tests

Test point. T4.1 – Validation of C_1. Before computing any optical spectrum, plot $C_1(T)$, $C_1(B)$, $\delta(T, B)$, and $\Delta_{\text{spin}}(T, B)$. Verify that they vary in the physically expected direction.

Test point. T4.2 – Chain size. Repeat the spin observables for several N_{spin}. The qualitative trends of C_1 and Δ_{spin} should be robust.

Test point. T4.3 – Boundary conditions. Compare open and periodic boundary conditions if the code allows it. If C_1 changes strongly, the optical interpretation must include finite-size uncertainty.

Test point. T4.4 – Optical control at fixed V_{BD}. Vary T and B in the spin chain, but force V_{BD} to remain constant in the optical model. If the spectrum still changes, there is another T, B-dependent channel that must be identified.

Test point. T4.5 – Active spin–optical control. Compare with the full case where $V_{\text{BD}}(T, B) = V_0 + \lambda_\delta \delta(T, B) + \lambda_C C_1(T, B)$. The difference between T4.4 and T4.5 isolates the spin–optical coupling effect.

Decision criterion. This step is the first real test of Q1 and Q7. If the spectral changes follow $C_1(T, B)$ or $\delta(T, B)$ monotonically and disappear when $\lambda_\delta = \lambda_C = 0$, then the model supports the idea that the optical response probes the spin sector.

10 Step 5 – Magnetic field: coherent mixing or simple energy shift

10.1 Objective

Question Q5 asks whether the magnetic field tunes coherent mixing or only shifts energies. We must therefore artificially decompose the channels through which B enters the model.

10.2 Scenarios

Scenario	What depends on B	Interpretation tested
B0	Nothing.	Numerical control.
B1	$T_{\mathrm{SP}}(B)$ and $\delta(T, B)$ only.	The field affects dimerization.
B2	$C_1(B)$ only.	The field affects magnetic correlations.
B3	Optical energies E_B, E_D only.	The field shifts resonances without changing mixing.
B4	Everything active.	Full model.

10.3 Tests

Test point. T5.1 – Shift versus intensity. For each scenario, extract (ω_1^*, ω_3^*), I_{cross}, I_{diag}, and R_{cross}. A pure energy shift should mainly change positions; a change in mixing should mainly change relative intensities and phases.

Test point. T5.2 – Cross-peak phase. The field may modify the phase even if the intensity changes little. Real and imaginary parts must therefore be retained, not only $|S|$.

Test point. T5.3 – Field derivatives. Compute

$$\frac{dI_{\mathrm{cross}}}{dB}, \qquad \frac{d\omega^*}{dB}, \qquad \frac{dR_{\mathrm{cross}}}{dB}. \tag{25}$$

These three derivatives distinguish coupling modulation from a simple spectral shift.

Decision criterion. If B mainly changes peak positions, the model indicates an energetic effect. If B mainly changes R_{cross}, phases, or the $R1/R2$ ratio, there is evidence for modulation of coherent mixing.

11 Step 6 – Temperature and Spin-Peierls transition

11.1 Objective

The Spin-Peierls transition provides a natural control: above T_{SP}, $\delta = 0$. If the signal attributed to dimerization persists unchanged above T_{SP}, the attribution is weak.

11.2 Tests

Test point. T6.1 – Scan around T_{SP}. Compute spectra for $T < T_{\mathrm{SP}}$, $T \simeq T_{\mathrm{SP}}$, and $T > T_{\mathrm{SP}}$. Extract I_{cross}, R_{cross}, and phases.

Test point. T6.2 – Separating δ from C_1. Above T_{SP}, $\delta = 0$ but C_1 may remain nonzero. This region tests whether the spectrum is sensitive to spin correlations without dimerization.

Test point. T6.3 – Control with $\lambda_C = 0$. If $\lambda_C = 0$, the spin-dependent signal should collapse with δ at the transition. If a signal persists, it comes from another mechanism.

Test point. T6.4 – Control with $\lambda_\delta = 0$. If $\lambda_\delta = 0$, the spectrum can remain sensitive to C_1 even above T_{SP}. This is the central test for whether short-range correlations are optically visible.

Decision criterion. This step answers Q2 and Q7. The strongest result would be a signature that clearly distinguishes long-range dimerization δ from short-range correlations C_1.

12 Step 7 – Add spin-sector dispersion

12.1 System

Activate:

$$N_k > 1, \qquad s = \texttt{spin\_mode\_dispersion\_scale} > 0, \qquad A_Q = 0. \tag{26}$$

The energies E_B and E_D should initially remain independent of k.

12.2 Objective

This step tests the idea that k mainly represents the spin/Spin-Peierls sector, not an optical d-band. It should show whether the dispersive spin dynamics leaves a measurable imprint on the 2D spectrum.

12.3 Tests

Test point. T7.1 – Convergence in N_k. Compute $N_k = 1, 5, 10, 25, 50, 100$. Integrated intensities must converge. If the structure changes strongly between 50 and 100, the model is not yet numerically stable.

Test point. T7.2 – Scan of s. Scan $s = 0, 0.25, 0.5, 1, 2$. The case $s = 0$ is the no-dispersion control.

Test point. T7.3 – k-resolved decomposition. Save or reconstruct $S_k(\omega_1, \omega_3)$ for several representative k values. This shows whether the integrated spectrum hides opposing contributions.

Test point. T7.4 – Keep optical energies flat. Keep $t_{B,r} = t_{D,r} = 0$. If optical dispersions are activated too early, one will no longer know whether changes come from spin or optical bands.

Decision criterion. Spin-sector dynamics can be called visible only if s modifies scalar observables or spectral shapes robustly, and if this effect disappears when $s = 0$.

13 Step 8 – Modulation of $g_Q(k)$

13.1 Objective

The parameter $A_Q =$ `g_Q_k_modulation` does not necessarily move the peaks. It redistributes the dynamic coupling over the k sector. One should therefore look for variations in intensity, spectral weight, and k-resolved contributions, not only for visual changes in normalized contours.

13.2 Tests

Test point. T8.1 – Non-normalized scan of A_Q. Use $A_Q = 0, 0.001, 0.005, 0.01, 0.05, 0.1, 0.25, 0.5, 0.75$. For each point, measure $I_{\max}$, I_{int}, I_{cross}, R_{cross}, and I_{R1}/I_{R2}.

Test point. T8.2 – Difference maps. Compute

$$\Delta S(A_Q) = S(A_Q) - S(0). \tag{27}$$

Difference maps must be non-normalized or normalized by a common scale.

Test point. T8.3 – Contributions by k region. Compare contributions near $k = 0$, $k = \pi/2$, and $k = \pi$. Since $g_Q(k) = g_Q(1 + A_Q \cos k)$, A_Q strengthens one region and weakens the other.

Test point. T8.4 – Sign control. Also test $A_Q < 0$. If the model is consistent, $A_Q > 0$ and $A_Q < 0$ should redistribute weight toward opposite k regions.

Decision criterion. If A_Q progressively decreases the total intensity without moving the peaks, this is not a failure: it is compatible with spectral-weight modulation. To interpret it physically, one must show which k region loses or gains weight.

14 Step 9 – Direct access to the dark state versus bright–dark mixing

14.1 Objective

The discussion of d-d transitions, selection rules, and the crystal field shows that two mechanisms must be distinguished:

$$\text{direct channel:} \quad \mu_D \neq 0, \quad V_0 = 0, \tag{28}$$

$$\text{mixing channel:} \quad \mu_D = 0, \quad V_0 \neq 0. \tag{29}$$

14.2 Tests

Test point. T9.1 – Scenario A: direct dark dipole. Set $V_0 = 0$, $\lambda_\delta = \lambda_C = 0$, but $\mu_D \neq 0$. Measure dark and cross peaks.

Test point. T9.2 – Scenario B: bright–dark mixing. Set $\mu_D = 0$, $V_0 \neq 0$. Adjust V_0 to obtain a dark intensity comparable to Scenario A.

Test point. T9.3 – Discrimination by pathways. Compare $R1$, $R2$, phases, and peak ratios. Two mechanisms that give the same linear intensity may give different 2D responses.

Test point. T9.4 – Temperature/field dependence. If the direct μ_D channel is purely crystal-field-like, it should not strongly follow $\delta(T, B)$ or $C_1(T, B)$, unless an additional hypothesis is introduced. The spin-dependent mixing channel should follow these variables.

Decision criterion. This step answers Q8 and clarifies the physical meaning of V_0. If Scenarios A and B are indistinguishable in the chosen observables, the model cannot conclude that the dark signal comes from SOC rather than from a weakly allowed direct dipole.

15 Step 10 – Thermal versus nonthermal states

15.1 Objective

Question Q6 requires changing the initial population without changing the Hamiltonian. Otherwise, one does not know whether the difference comes from a nonthermal state or from a parameter change.

15.2 Tests

Test point. T10.1 – Thermal reference state. Use Boltzmann populations for a temperature T.

Test point. T10.2 – Simple nonthermal population. Build a distribution in which selected bright/dark levels or selected mode levels are overpopulated. Keep the same Hamiltonian and the same dipoles.

Test point. T10.3 – Comparison at equal average energy. Compare a thermal state and a nonthermal state with similar average energy. If the spectra differ, spectroscopy is sensitive to the population structure, not only to energy.

Test point. T10.4 – t_2 dependence. If the code allows a population-time scan t_2, look for oscillations or dephasings that differ between the thermal and nonthermal states.

Decision criterion. Sensitivity to nonthermal states can be claimed only if two states with identical Hamiltonian parameters produce distinct and robust 2D signatures.

16 Step 11 – Robustness and artifacts

16.1 Objective

Before drawing conclusions, one must show that the trends do not depend on an arbitrary numerical choice.

16.2 Tests

Test point. T11.1 – Robustness to η. Trends in I_{cross}, R_{cross}, and peak positions must survive reasonable variations of the broadening.

Test point. T11.2 – Grid robustness. Repeat the main results with a finer frequency grid and a slightly wider spectral window.

Test point. T11.3 – Robustness to N_k. Conclusions about A_Q, s, and $\omega_Q(k)$ must remain stable when N_k is increased.

Test point. T11.4 – Robustness to chain size. Conclusions related to C_1 and Δ_{spin} must be compared for several chain sizes.

Test point. T11.5 – Robustness to integration windows. The integrals I_{cross} and I_{diag} must be recomputed with slightly different windows.

Decision criterion. A scientific conclusion should be classified as robust only if it survives at least three controls: no panel normalization, numerical convergence, and a physical control in which the mechanism is switched off.

17 Step 12 – Extension toward explicit SOC

17.1 Why not start there

An explicit Hamiltonian

$$H_{\mathrm{SOC}} = \lambda \boldsymbol{L} \cdot \boldsymbol{S} \tag{30}$$

is more physical, but it increases the Hilbert-space size and introduces new choices: L_α matrices, orbital basis, dipoles, crystal field, and possible double counting with V_{BD}.

17.2 Route 1: effective spinor model

The first reasonable extension is:

$$(g, D, B) \otimes (\uparrow, \downarrow) \otimes \mathcal{H}_{\mathrm{ph}}. \tag{31}$$

For $n_{\mathrm{bosons}} = 3$, the Hilbert dimension per k increases from 9 to 18, and the Liouville dimension from 81 to 324.

Test point. T12.1 – Recover the effective model. Choose simple L_α^{eff} matrices and verify that, at weak λ_{eff}, the signatures resemble the model with V_{BD}.

Test point. T12.2 – Avoid double counting. When H_{SOC} is active, replace V_{BD} by a residual V_{res}, or set it to zero in the first test.

Test point. T12.3 – Spin-flip signatures. Verify whether new pathways appear when S_x and S_y are active. These pathways would be absent from a scalar bright–dark model.

17.3 Route 2: microscopic d-orbital model

The more complete route is:

$$\{d_{x^2-y^2}, d_{xy}, d_{xz}, d_{yz}, d_{z^2}\} \otimes (\uparrow, \downarrow) \otimes \mathcal{H}_{\mathrm{ph}}. \tag{32}$$

For $n_{\mathrm{bosons}} = 3$, the Hilbert dimension per k is about 30, and the Liouville dimension about 900.

Decision criterion. Route 2 should be undertaken only after identifying, with Route 1, which 2D observable is truly sensitive to explicit spin. Otherwise, the microscopic model risks adding parameters without increasing the strength of the conclusion.

18 Recommended calculation order

Priority	Calculation	Why	Target question
1	V_{BD} scan, $\mu_D = 0$, $g_Q = 0$, $N_k = 1$.	Prove that cross peaks come from mixing.	Q3
2	g_Q scan, $N_k = 1$.	Separate static mixing from dynamic mode coupling.	Q1, Q3
3	Manual scans of δ, C_1.	Determine whether the spectrum measures only V_{BD} or distinguishes its components.	Q4, Q7
4	Finite spin chain: $C_1(T, B)$, $\delta(T, B)$.	Connect the optical model to spin observables.	Q1, Q7
5	Temperature scan around T_{SP}.	Use the transition as a physical control.	Q2, Q7
6	Magnetic-field decomposition.	Distinguish energetic shift from mixing change.	Q5
7	Activate $\omega_Q(k)$, $A_Q = 0$.	Test dispersive spin dynamics.	Q1
8	Scan A_Q.	Test redistribution of coupling in k.	Q1, Q9
9	Compare $\mu_D \neq 0$ versus $V_0 \neq 0$.	Distinguish direct dipole from hybridization.	Q8
10	Nonthermal states.	Test sensitivity to populations.	Q6
11	Effective spinor model.	Replace V_{BD} by explicit SOC.	Q1–Q8

19 Conclusion criteria

19.1 Strong conclusion

One can strongly support the claim that multidimensional spectroscopy probes the spin sector if the following conditions are satisfied:

1. Cross peaks disappear when bright–dark mixing and the direct dark channel are both switched off.
2. Cross peaks systematically follow δ, C_1, or $\omega_Q(k)$ when these quantities are activated separately.
3. The trends persist without panel normalization.
4. The trends converge in grid, N_k, η, and chain size.
5. A control with $\lambda_\delta = \lambda_C = g_Q = 0$ removes the spin-dependent signature.

19.2 Moderate conclusion

The conclusion will be moderate if the spectra are sensitive to V_{BD}, but do not separate δ and C_1. In that case, the model shows that spectroscopy can probe an effective mixing parameter, but cannot yet identify its microscopic source.

19.3 Weak or negative conclusion

The conclusion will be weak if:

1. the same signatures appear with $\mu_D \neq 0$ and without spin coupling;
2. the signatures disappear under small changes of normalization or grid;
3. field or temperature variations only shift peaks without changing cross-peak ratios;
4. effects assigned to k do not converge with N_k.

20 Recommended deliverables

For each important step, save:

1. raw spectra in numerical format;
2. non-normalized figures;
3. normalized figures only for visual comparison;
4. a complete parameter file;
5. a CSV table of scalar observables;
6. a short note indicating which mechanism was active or inactive.

The minimal observable table should contain:

$$I_{\max}, \quad I_{\text{int}}, \quad I_{\text{cross}}, \quad I_{\text{diag}}, \quad R_{\text{cross}}, \quad I_{R1}, \quad I_{R2}, \quad \omega_1^*, \quad \omega_3^*. \tag{33}$$

21 Next concrete action

The next calculation should be Step 1:

$$N_k = 1, \qquad g_Q = 0, \qquad \mu_D = 0, \qquad V_{\text{BD}} \text{ manually scanned.} \tag{34}$$

This calculation is the smallest possible test of the core of the model. It answers the question:

Are the cross peaks that we observe truly generated by coherent bright–dark mixing?

Without this validation, it would be premature to interpret scans in $g_Q(k)$, δ, C_1, T, or B as signatures of the spin sector.